

Debasish Chakroborti (Joy), Ph.D.

✉ debasish.chakroborti@usask.ca 🐦 @joydeba 🏠 410 Stensrud Rd #118, Saskatoon 📞 306-914-6212 🌐
<https://joydeba.github.io/> 📄 www.stackoverflow.com/users/1797926/joydeba 🌐 www.github.com/joydeba in
www.linkedin.com/in/joydeba/ 🎓 <https://www.researchgate.net/profile/Debasish-Chakroborti>

Employment History and Projects

- 2024-... ■ **Faculty at the School of Computing and Digital Innovation, Saskatchewan Polytechnic** - Conducted classes in Computer System, Cloud Computing and Blockchain Program.
- **Living Skies Postdoctoral Fellow, UofS** - NSERC CREATE Software Analytics Research (SOAR) project.
- 2025-26 ■ **Consultant (AI Automation Team), Cloud Linux Software, Inc., a Delaware Corporation, USA.** - Provided expert advice on developing an AI-based automated backporting tool.
- 2024-24 ■ **Sessional Lecturer, Department of Computer Science, University of Saskatchewan.** - Delivered lectures and conducted labs on topics in Computer Science.
- **Software Engineer (Intern), Push Interactions.** - Enhanced <https://www.wilger.net> website.
- 2023-23 ■ **PhD Intern (Remote), Synesis IT GmbH, Germany.** - Conducted API planning and managed big data databases for Traffic Management **Intelligent Transportation Systems**.
- 2022-24 ■ **GSA Office Coordinator (Part-time), Graduate Students' Association.** - Maintained the **GSA website** at <https://gsa.usask.ca>, added new web pages based on services and events. HTML and JavaScript.
- **Software Engineer (Part-time), Saskatchewan Education Leadership Unit.** - Maintained the **Placement Application** at <https://placement.usask.ca>. Developed a matching model for Teacher Candidates and Mentors.
- 2017-24 ■ **TA, RA, Masters and, PhD, UofS, Plant Phenotyping and Imaging Research Centre, Global Institute for Food Security.** - Developed **RISP and RISPs** data re-usability algorithms within WfM System (Hadoop and Spark clusters). Developed tools in Software Engineering and Image Processing domains such as **Backzer, ReBack (CNN), BackTrans (LLM)** and **BackSlice** for analyzing, recommending, and adapting backporting pull-requests and **InsCount** (based on UNet) for counting dense organs in plant images.
- 2013-24 ■ **Lecturer, Computer Science and Engineering Discipline, Khulna University, Bangladesh.** - Conducted classes and research, and implemented the websites for Khulna University (ku.ac.bd) and ICCIT (iccit.org.bd/2013/). Html, css, JavaScript, R, Python, Java, .Net, Apache Web Server, Github.
- 2013-13 ■ **Software Engineer, Nascenia Limited, Bangladesh.** - **SnapKnot.** Developed a centralized wedding photographer directory website (<https://snapknot.com/>). Implemented backend functionality for searching photographers and couples using Ruby on Rails, ASP.NET, C#, Git.
- 2012-13 ■ **Software Engineer, Divine IT Limited, Bangladesh.** - Worked on various ERP systems (<https://www.divineit.net>) including Lines Pay, Sphere, and Prism. Django, Mysql, Postgresql, sqlite, SVN.

Activities and Achievements - Awards, Volunteering

- 2017-23 ■ **Over ten scholarly articles in renowned journals and conferences, including ICSE, ICPC, MSR, JSME, EMSE, EICS, Big Data, and ASE.**
- 2008-23 ■ **Carl McCrosky Innovation Scholarship for Graduate Studies 2023, GSA Mark Kroecker Exceptional Student Leadership Award 2023, Saskatchewan Innovation and Opportunity Scholarship, Ph.D. Citizenship Award 2024, Departmental Scholarship, SOAR, SEMLA, CSGC Best Poster Awards, Ivan and Margaret Toutloff Award, Graduate Travel Awards for Seattle-Melbourne- Montreal Visits, Innovation Award Khulna, Deans Merit Scholarship Khulna University.**
- 2017-22 ■ **President** of the Computer Science Council, **Volunteer Lead** at **Digitized-Technology** event for Saskatchewan students, **Vice President-Social, Vice President-Admin** of the CSGC-UofS, **Member** of Research Fest Committee-UofS, **Vice President - Events** of the Bangladeshi Association-Usask, **Volunteer Lead** at SSPP, **Volunteer** at **Global village**, **Volunteer** at **Folk-fest**, **GSA rep** for the **Associate Dean Students Search Committee**, **GSA rep** on **Arts and Science Planning Advisory Committee**.

Education

- 2020-24 ■ **Ph.D., Computer Science, University of Saskatchewan, Canada**
- 2017-19 ■ **M.Sc., Computer Science, University of Saskatchewan, Canada**
- 2008-12 ■ **B.Sc., Computer Science and Engineering, Khulna University, Bangladesh.**

Research Publications

Journal Articles

- 1 **Chakroborti, D.**, Schneider, K., & Roy, C. (2024). Reback: Recommending backports in social coding environments. *Automated Software Engineering*, 31(1), 18. [doi:10.1007/s10515-024-00416-1](https://doi.org/10.1007/s10515-024-00416-1)
- 2 **Chakroborti, D.**, Nath, S., Schneider, K., & Roy, C. (2022). Release conventions of open-source software: An exploratory study. *Journal of Software: Evolution and Process*, 05, JSME-22, e2499. [doi:https://doi.org/10.1002/smr.2499](https://doi.org/10.1002/smr.2499)
- 3 **Chakroborti, D.**, Roy, B., & Nath, S. S. (2021). Designing for recommending intermediate states in a scientific workflow management system. *Proc. ACM Hum.-Comput. Interact.*, 5(EICS). [doi:10.1145/3457145](https://doi.org/10.1145/3457145)
- 4 Herbold, S., Trautsch, A., Ledel, B., Aghamohammadi, A., Ghaleb, T., Chahal, K., ... **Chakroborti, D.** (2021). A fine-grained data set and analysis of tangling in bug fixing commits. *Empirical Software Engineering*.
- 5 Bairagi, A. K., Mondal, S., & **Chakroborti, D.** (2017). Securing bangla text communication using image steganography with dynamic substitution in iot environment. *Khulna University Studies, Khulna University, Bangladesh*, 14, (KUS: 16/30:010816).
- 6 **Chakroborti, D.**, Roy, C., & Schneider, K. (ob). Code propagation in stable version management: A systematic literature review. Manuscript under review.

Conference Proceedings

- 1 Tasnim, J., **Chakroborti, D.**, Roy, C., & Schneider, K. (2025). An insight into the technical debt-fix trade-off in software backporting. In *Proceedings of the 35th ieee international conference on collaborative advances in software and computing (cascon 2025)*, York University, Toronto, ON, Canada: IEEE.
- 2 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2024). A study of backporting code in open-source software for characterizing changesets. In *2024 ieee/acm 46th international conference on software engineering: Companion proceedings (icse-companion '24)*. [doi:10.1145/3639478.3643079](https://doi.org/10.1145/3639478.3643079)
- 3 Tasnim, J., **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2023). How does quality deviate in stable releases by backporting? In *45th international conference on software engineering (icse 2023) at: May 14-20, melbourne, australia*.
- 4 **Chakroborti, D.**, Schneider, K. A., & Roy, C. K. (2022). Backports: Change types, challenges and strategies. In *30th ieee/acm international conference on program comprehension (icpc 2022)at: May 16-17, pittsburgh, pa, usa*. [doi:10.1145/3524610.3527920](https://doi.org/10.1145/3524610.3527920)
- 5 Bhattacharjee, A., Nath, S. S., Zhou, S., **Chakroborti, D.**, Roy, B., Roy, C. K., & Schneider, K. (2020). An exploratory study to find motives behind cross-platform forks from software heritage dataset. In *Proceedings of the 17th international conference on mining software repositories* (pp. 11–15). [doi:10.1145/3379597.3387512](https://doi.org/10.1145/3379597.3387512)
- 6 **Chakroborti, D.**, Mondal, M., Roy, B., Roy, C. K., & Schneider, K. A. (2018). Optimized storing of workflow outputs through mining association rules. In *2018 ieee international conference on big data (big data)* (pp. 508–515). [doi:10.1109/BigData.2018.8622351](https://doi.org/10.1109/BigData.2018.8622351)
- 7 **Chakroborti, D.**, & Nath, S. S. (2017). Web service performance enhancement for portable devices modifying soap security principle. In *2017 20th international conference of computer and information technology (iccit)* (pp. 1–7). [doi:10.1109/ICCI TECHN.2017.8281799](https://doi.org/10.1109/ICCI TECHN.2017.8281799)
- 8 Bairagi, A. K., & **Chakroborti, D.** (2015). Trust based d2d communications for accessing services in internet of things. In *2015 18th international conference on computer and information technology (iccit)* (pp. 50–54). [doi:10.1109/ICCI TECHN.2015.7488041](https://doi.org/10.1109/ICCI TECHN.2015.7488041)
- 9 **Chakroborti, D.**, Roy, C., & Schneider, K. (oa). Backslice: Achieving accurate and essential propagation through backport slicing. Submitted.

Book Chapters and Posters

- 1 **Chakroborti, D.**, Roy, C. K., & Schneider, K. A. (2022). **BEST POSTER (3rd):** Backporting for version management: Automated changesets integration in software repositories. *symposium on software engineering for machine learning applications*.
- 2 Tasnim, J., **Chakroborti, D.**, Roy, C. K., Schneider, K. A., & D., M. (2022). **BEST POSTER (2nd):** Detecting and visualizing quality aspects in backporting pull requests. *symposium on innovations in computer science and applied computing (icsac)*.

- ③ **Chakroborti, D.,** Roy, C. K., & Schneider, K. A. (2021). ***BEST POSTER:** Identifying and integrating changesets in backporting. the annual software analytics research (soar) symposium.*
- ④ **Chakroborti, D.,** Roy, B., Mondal, A., Mostaeen, G., Roy, C. K., Schneider, K. A., & Deters, R. (2020). *A data management scheme for micro-level modular computation-intensive programs in big data platforms* (R. Alhajj, M. Moshirpour, & B. Far, Eds.). [doi:10.1007/978-3-030-32587-9_9](https://doi.org/10.1007/978-3-030-32587-9_9)